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A Transformer Network for Settlement Generalization Integrating Raster Maps and Remote Sensing Images

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Highlights

What are the main findings?

- A DSG-Trans model integrating raster maps and remote sensing imagery was developed for intelligent multi-scale settlement generalization.
- An effective scale-matching relationship was identified between target map scales and remote sensing image resolutions, and appropriate resolution selection significantly improved settlement generalization performance.

What are the implications of the main findings?

- Specific low-resolution remote sensing imagery was demonstrated to provide scale-guidance information for map generalization, offering a new theoretical perspective for intelligent map generalization.
- A map generalization framework integrating data-driven learning and scale constraints was established, providing a new technical pathway for automated map production and multi-scale spatial information representation.

Abstract

To address the challenges of reliance on manual expertise and insufficient automation and intelligence in settlement generalization during scale transformation, this paper proposes a multi-scale settlement generalization model (DSG-Trans) based on the fusion of dual-source data: raster maps and remote sensing images. Leveraging the cartographic correspondence between remote sensing image resolution and map scale, this model integrates source-scale maps with specific low-resolution remote sensing imagery to automatically generate target-scale raster maps that adhere to cartographic specifications. The model employs a bidirectional cross-attention mechanism and a Transformer architecture to facilitate the semantic integration of dual-source data, while a progressive upsampling strategy is utilized to reconstruct spatial details. Additionally, a composite loss function incorporating four constraints is designed to enhance the quality of settlement generalization. Experiments were conducted in New York City, USA, using a multi-source dataset constructed from 3D-GloBFP building footprint data and multi-source remote sensing imagery, focusing on two scale transformation scenarios: 1:5000 to 1:10,000 and 1:5000 to 1:25,000. The results indicate that the proposed method effectively learns and simulates expert generalization knowledge, significantly improving the automation level and scale adaptability of map generalization while preserving geometric morphology and spatial distribution patterns.

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1. Introduction

Map generalization serves as the fundamental technique for the multi-scale representation of geographic features; it functions as a cartographic method that reconstructs geographical patterns appropriate to a target scale through operators such as selection, simplification, aggregation, and relation coordination, constrained by map purpose, scale, and regional geographic characteristics. As pivotal carriers characterizing urban spatial structure, settlements have long remained a focal point in map generalization research, where the quality of settlement generalization directly influences the visual legibility, aesthetic quality, and the accuracy of subsequent spatial analyses derived from cartographic products. Settlement generalization is a complex cognitive and decision-making process that entails not only the application of operators such as selection, simplification, amalgamation, and displacement, but also the preservation of shape similarity among building polygons and the consistency of spatial distribution patterns within building groups, while strictly maintaining typical morphological features of artificial objects such as the parallelism and orthogonality of building polygon edges.

Previous studies on settlement generalization have predominantly been limited to single map data sources. In the era of big data, the fusion of multi-source geospatial data has emerged as a critical approach within spatial information science, effectively mitigating the limitations of information incompleteness inherent in single data sources [1]. Nevertheless, determining which spatial data to integrate with map data to support map generalization requires further investigation [2]. At present, commercial companies like Google and Microsoft widely utilize remote sensing images to produce global vector building footprints [3], whose resolution is vital for the scale and cartographic accuracy of the generated vector data. The theoretical basis of this paper is established on the cartographic correspondence between the resolution of remote sensing imagery and map scale. Maps represent the outcome of the selection and generalization of surface features at specific scales, serving as a synthesis of human subjectivity and objective geographic reality, where physical entities are mapped to cartographic objects through abstract symbology. By contrast, remote sensing imagery constitutes a physical recording of the Earth's surface by sensors under specific observational conditions; it provides a direct reflection of the objective world, discretizing the continuous surface into raster cells, the dimensions of which define its spatial scale. In large-scale topographic maps, settlements are depicted as discrete individual features. As the map scale decreases, these settlements are amalgamated into blocks to accommodate the constraints of the reduced scale. Correspondingly, as the spatial resolution of remote sensing imagery decreases, the detailed contours of settlements become gradually blurred, and the boundaries between individual features become increasingly indistinguishable. This variation in human visual perception is somewhat similar to operators like simplification, aggregation, and selection in settlement generalization. Motivated by this, we propose that employing lower-resolution remote sensing images aligned with the target scale as a preset scale guide can constrain the generalization degree of map features, enhancing the scale adaptability of cartographic models.

This paper presents a settlement generalization method that integrates maps and remote sensing images. The proposed method uses large-scale settlement map data and specific low-resolution remote sensing imagery as dual data sources, and introduces the **DSG-Trans** model (**D**ual-source **S**ettlement **G**eneralization **Trans**former) with the rep-

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resentative Transformer [4] architecture serving as the core module. This enables the learning of map generalization knowledge from high-quality map samples to perform settlement generalization from large to small scales. Figure 1 illustrates the implementation workflow of the proposed method. Specifically, the source map offers geometric details and semantic attributes, whereas the remote sensing imagery supplies scale and context information. Furthermore, unlike Convolutional Neural Networks (CNNs) constrained by local receptive fields, the Transformer can precisely capture long-range spatial relations between settlement objects, showing notable advantages for settlement generalization tasks requiring context information.

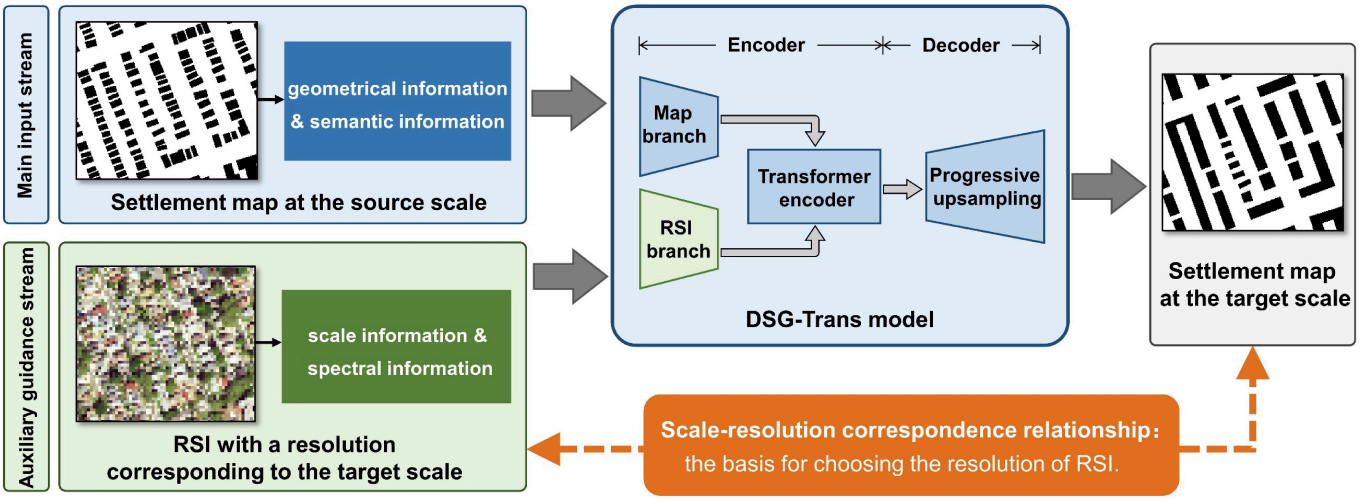

Figure 1. The implementation workflow of the proposed method.

The primary contributions of this paper are summarized as follows:

1. Conceptually, we propose a new map generalization paradigm that uses the scale information from multi-resolution remote sensing images as an auxiliary guide, which focuses on mining the visual characteristics of low-resolution images akin to boundary simplification and feature aggregation after generalization. However, previous remote sensing cartography studies have predominantly focused on using the location and shape features of geo-objects derived from single-resolution imagery.
2. Methodologically, an end-to-end DSG-Trans model is established. A bidirectional cross-attention mechanism is introduced to bridge the information gap between maps and images; meanwhile, the Transformer architecture overcomes the local receptive field limits of traditional CNNs and achieves deep semantic integration of multi-source heterogeneous data, remarkably enhancing the quality of settlement generalization.
3. Through a series of experiments, the matching relationship between remote sensing image resolution and the target scale is investigated to validate the effectiveness of images as scale guidance.

The rest of this paper is organized as follows: Section 2 reviews traditional and deep learning approaches to settlement generalization; Section 3 details the implementation of the dual-source fusion strategy for settlement generalization, covering problem definition, DSG-Trans model architecture, and loss functions; Section 4 presents the experimental results and analysis; Section 5 offers a discussion of the proposed method; and Section 6 concludes this paper.

2. Related Work

2.1. Traditional Settlement Generalization Methods

Traditional settlement generalization predominantly relies on explicit cartographic rules and constraints to simulate the manual generalization process. Numerous algorithms have been developed specifically for various generalization operators, including selection, simplification, amalgamation, and displacement. Among these, the selection operator is applicable to settlements that degenerate into point features on small-scale maps, where the quota method and qualification method are frequently employed to determine the quantity and criteria for selection [5].

Polygon simplification and amalgamation represent common generalization operators for areal settlements at large scales. Representative simplification methods include the following: Regnauld et al. achieved simplification by eliminating small protrusions and depressions [6]; Sester realized shape simplification by fitting multiple vertices to straight lines using a least squares adjustment method subject to constraints [7]; additionally, there are template matching methods that replace objects with similar shapes matched from a template library [8], as well as image processing methods based on logical operations on raster data [9]. These methods effectively preserve the parallelism and orthogonality characteristics of settlement boundaries.

When the spatial interval between buildings in the map falls below a cartographically appropriate threshold, amalgamation of polygons becomes necessary. Amalgamation entails considering both polygon grouping (i.e., which polygons need to be merged) and merging methods (i.e., ensuring that the post-amalgamation shape maintains the consistency of spatial distribution patterns before and after generalization). These algorithms are generally categorized into raster-based methods and vector-based methods. Raster-based methods primarily implement settlement amalgamation through mathematical morphological operations and their combinations [10,11]. For instance, Su et al. constructed an algebraic model utilizing dilation, erosion, and closing operations, decomposing the process into two stages—natural amalgamation and shape refinement—to achieve automatic generalization [12]. Vector-based methods predominantly utilize Delaunay triangulation to establish spatial proximity relationships among polygons as the basis for amalgamation [13,14]. For example, Ai et al. employed constrained Delaunay triangulation to detect the gap morphology and adjacency characteristics between settlements, proposing three amalgamation modes: absorption, encapsulation, and decomposition [15].

Due to the influence of factors like non-scale and semi-scale map symbols, spatial conflicts of map features often arise, necessitating the displacement algorithms to adjust feature positions. Displacement methods are generally classified into two categories: the first involves iterative conflict detection and gradual movement until all conflicts are resolved, often employing optimization algorithms such as simulated annealing [16] and genetic algorithms [17]; the second computes the displacement vector (distance and direction) in a single step based on an analysis of the contextual environment of map objects within the conflict zone, utilizing approaches such as triangulation [18] and field theory [19].

Given the complexity of the map generalization process, it is challenging to orchestrate the selection, simplification, amalgamation, and displacement of building polygons using a limited set of rules or models [20]. Furthermore, the application of these algorithms usually relies on expert experience to choose appropriate algorithms and parameters. Consequently, traditional methods are suitable for human–computer interaction environments, but it is still difficult to achieve true automation and intelligence in settlement generalization.

2.2. Intelligent Settlement Generalization Methods Based on Deep Learning

The rapid advancement of Deep Learning (DL) has introduced a new paradigm to map generalization [21], enabling the direct acquisition of generalization rules and domain knowledge from massive map datasets while demonstrating superior generalization capability and computational efficiency. The map generalization process encompasses four key sub-tasks: knowledge acquisition, algorithm design, process control, and quality evaluation. However, current research on intelligent settlement generalization primarily focuses on knowledge acquisition and algorithm design [22].

The acquisition of settlement generalization knowledge serves as the prerequisite and foundation for implementing settlement generalization, encompassing the shape recognition of individual buildings and the distribution pattern recognition of building groups. Liu et al. and Hu et al. directly employed the vertex features of buildings as inputs for DL models to perform shape recognition [23,24]; Yan et al. modeled buildings using graph structures and constructed a Graph Convolutional Autoencoder (GCAE), achieving shape encoding and recognition through unsupervised feature aggregation [25]. In terms of building group pattern recognition, Zhao et al. proposed a Graph Convolutional Network (GCN) integrated with a spectral Transformer network module, which is capable of automatically identifying various forms of building group patterns [26].

In research on map generalization algorithms, a prevailing approach treats settlement generalization as an image-to-image translation task, utilizing visual models such as CNNs and Generative Adversarial Networks (GANs) to learn the mapping relationships between pre- and post-generalization images [27,28]. For example, Sester et al. applied ResU-Net to raster maps to achieve end-to-end settlement generalization [29]; Zhou et al. explicitly introduced spatial awareness into DL models, proposing a Spatially aware GAN model (SpaGAN) [30]. Given the analogy between the encoding–decoding process and the map generalization process, these models commonly employ deep neural networks with an encoder–decoder architecture [31]. Although raster-based methods may suffer from issues such as noise and distortion, they can naturally coordinate multiple generalization operators and achieve end-to-end solutions. There are also several studies that directly construct settlement representations from vector maps and learn generalization knowledge, typically modeling individual buildings or building groups as graph structures, and converting operations such as selection, simplification and amalgamation into GCN-based classification or regression tasks [32,33]. Zhou et al. proposed a multi-task GCN (MT_GCN) that decomposes simplification into two sub-tasks, namely vertex removal and displacement, and achieves settlement simplification through the joint learning of classification and regression [34]. Yan and Yang developed a settlement simplification method based on a graph autoencoder that allows for the flexible configuration of spatial constraints [35].

Within the domain of map generalization, a few studies have drawn inspiration from the sequence generation paradigm of Transformers, framing settlement simplification as a transformation problem of vector coordinate sequences [36–38]. The most prevalent application scenario of the Transformer model within the GIS domain lies in remote sensing imagery interpretation and mapping, specifically in extracting vector settlements from imagery to serve as cartographic inputs [39,40]. Such research primarily focuses on mining the location and shape information of features in remote sensing imagery, representing a "static" interpretation task of feature contours within single-resolution imagery, while the exploitation of scale information remains limited. Unlike the aforementioned tasks, this paper addresses the problem of "dynamic" map generalization under scale transformation. By mining scale information from multi-resolution remote sensing imagery, the proposed approach facilitates the derivation of multiple small-scale maps from large-scale maps, satisfying the representation requirements of the target scale.

3. Methodology

3.1. Problem Definition

In the proposed method, maps and remote sensing images are complementary in information representation: maps provide precise geometry, topological relations, and semantic attributes but lack the context information required at the target scale; conversely, low-resolution remote sensing images, whose visual effects stem from mixed pixels, cannot represent topology and attributes but can reflect the scale and context information at the corresponding level. Therefore, this paper treats remote sensing images as scale-aware auxiliary information, which collaboratively drives the generalization process alongside maps.

Based on the above understanding, this paper formulates settlement generalization as a binary image mapping problem guided by remote sensing image data. Let the source raster map be $X_{Map} \in R^{m \times m \times 1}$, the remote sensing imagery matching the target scale be $X_{RSI} \in \mathbb{R}^{r \times r \times 3}$, and the target raster map be $Y \in \mathbb{R}^{m \times m \times 1}$ (m and r denote the sizes of image tiles); thus, the training sample set $\mathcal{D} = \left\{ (X_{Map}^{(i)}, X_{RSI}^{(i)}, Y^{(i)}) \right\}_{i=1}^{N}$ can be established. Here, the binary raster map is generated by rasterizing vector settlement data, with each pixel labeled as either settlements (value 1) or background (value 0). This paper aims to learn a parameterized function $\mathcal{G}_\theta$ (i.e., the DSG-Trans model) that maps the dual-source inputs to the target-scale output, as expressed in Equation (1):

$$\hat{Y}^{(i)} = \mathcal{G}_\theta \left(X_{Map}^{(i)}, X_{RSI}^{(i)} \right) \tag{1}$$

where $\hat{Y}^{(i)}$ denotes the probability map generated by the model with pixel values in the range of [0, 1], representing the probability of each pixel belonging to settlements, and θ denotes all the learnable parameters of the network. During the training phase, the network parameters θ are iteratively optimized by minimizing the loss function $\mathcal{L}$ between the predicted probability map $\hat{Y}^{(i)}$ and the ground-truth target map $Y^{(i)}$. Ultimately, the probability map output by the trained model is binarized and regularized before being converted into standard vector data.

3.2. The Dual-Source Settlement Generalization Transformer (DSG-Trans) Model

As illustrated in Figure 2, the DSG-Trans model adopts an Encoder–Decoder architecture. The encoder primarily includes the Parallel CNN Branch (PCB), the Bidirectional Cross-Attention Fusion Module (BCAFM), and the Transformer Encoder Block, while the decoder is composed of multiple cascaded upsampling modules. The PCB initially receives the source map and remote sensing imagery, and the extracted features are subsequently serialized and fed into the BCAFM for semantic alignment, followed by deep feature integration via the Transformer Encoder Block. The decoder accepts Latent Features and multi-level Skip Connection Features, reconstructing settlement spatial details through a progressive upsampling strategy to ultimately generate the target map. The overall computational procedure of DSG-Trans is summarized in Algorithm 1.

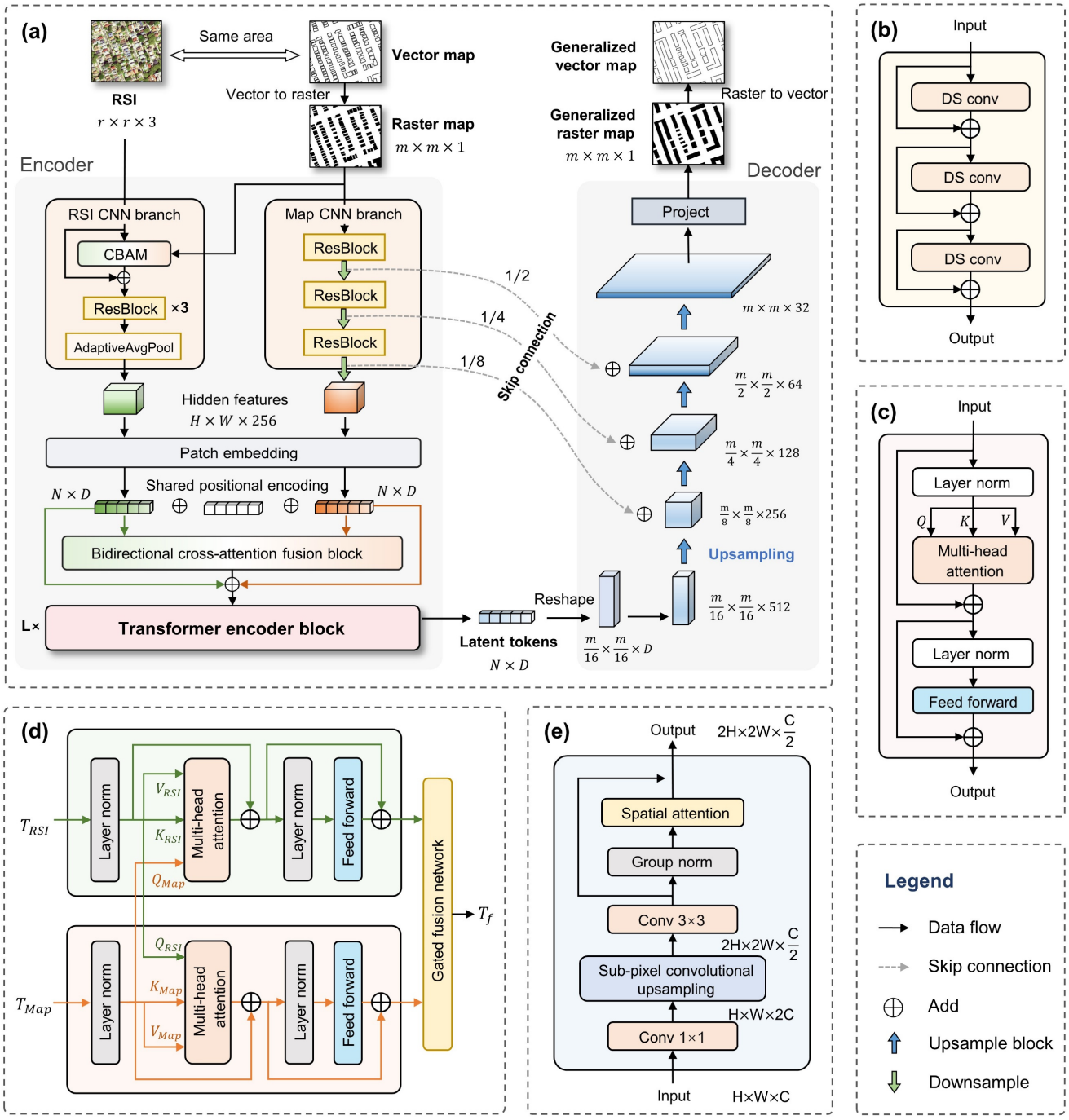

Figure 2. Structure of the DSG-Trans model and its constituent blocks. (**a**) DSG-Trans; (**b**) ResBlock; (**c**) Transformer Encoder Block. (**d**) Bidirectional Cross-Attention Fusion Module (BCAFM). (**e**) Upsampling Block, where channel dimensions are adjusted prior to input into the sub-pixel convolutional layer.

3.2.1. Dual-Branch Encoder Based on CNN and Transformer

1. Parallel CNN Branches (PCB).

The PCB is designed for feature extraction, where patch embedding is applied to its output hidden features rather than the raw images. This design is adopted for two reasons: first, the low-resolution, high-semantic feature maps extracted by CNNs facilitate feature fusion; second, the multi-level convolutional outputs provide hierarchical spatial details for the decoder.

As illustrated in Figure 2a,b, the PCB extracts hidden features from the map and remote sensing imagery via the Map Branch and the RSI Branch, respectively. Each branch is primarily composed of Residual Blocks (ResBlocks) based on Depthwise Separable Convolutions (DS Conv). The ResBlock in the RSI Branch maintains a constant feature size, aligning the dimensions solely through an Adaptive Average Pooling (AdaptiveAvgPool) layer at the end of the branch. Meanwhile, the ResBlock in the Map Branch sets the stride of the last DS Conv to 2 for multi-level downsampling. The multi-resolution shallow features obtained from multi-level downsampling serve as Skip Connection Features, which are combined via a weighted sum with the corresponding upsampling features in the decoder. To suppress interference from background pixels in the remote sensing imagery, the model concatenates the two types of data along the channel dimension and inputs them into CBAM [41] for feature calibration. Ultimately, the PCB outputs two hidden features, H_{Map} and $H_{RSI} \in \mathbb{R}^{H \times W \times 256}$, with the same shape.

Algorithm 1. DSG-Trans-based dual-source settlement generalization.

Input: Training set $\mathcal{D} = \{(X_{Map}^{(i)}, X_{RSI}^{(i)}, Y^{(i)})\}_{i=1}^{N}$; DSG-Trans model $\mathcal{G}_\theta$

Output: Target-scale settlement probability map $\hat{Y}^{(i)}$ and generalized settlement data

for each sample $(X_{Map}^{(i)}, X_{RSI}^{(i)}, Y^{(i)}) \in \mathcal{D}$ **do**

  Extract Map and RSI features using the PCB, and retain multi-level map features for skip connections;

  Calibrate the dual-source features using CBAM to suppress background interference from X_{RSI}^{i};

  Serialize the hidden features through patch embedding and perform bi-directional feature interaction using BCAFM;

  Fuse the interacted features through the Gated Fusion Network;

  Feed the fused sequence into the Transformer Encoder Block to model long-range spatial dependencies;

  Reconstruct settlement spatial details through the Progressive Upsampling Decoder with skip connections;

  Generate the target-scale probability map $\hat{Y}^{(i)} = \mathcal{G}_\theta(X_{Map}^{(i)}, X_{RSI}^{(i)})$;

  During training, calculate $\mathcal{L}(\hat{Y}^{(i)}, Y^{(i)})$ and update θ by backpropagation;

end for

During inference, binarize the predicted probability map and perform vectorization and regularization to obtain the final generalized settlement data.

2. Bidirectional Cross-Attention Fusion Module (BCAFM).

The hidden features H_{Map} and H_{RSI} are split into N patches of shape $P \times P \times 256$ (where P is the patch size, $N = HW/P^2$), and each patch is then mapped into a 1D vector to generate two sets of sequences. Subsequently, shared learnable position encodings are applied to these sequences to ensure the spatial alignment of the multi-source data. This process yields the representations T_{Map}, $T_{RSI} \in \mathbb{R}^{N \times D}$ (where D represents the embedding dimension).

The cross-attention mechanism is an effective means for multi-source information fusion [42,43]. As illustrated in Figure 2d, the BCAFM achieves information interaction and alignment between the map and remote sensing imagery by exchanging *Query* vectors within the Multi-Head Attention (MHA) structure. Specifically, when T_{Map} provides the

Query, it is equivalent to the map retrieving texture details matching the target scale from remote sensing imagery, thereby determining feature levels to be preserved; when T_{RSI} provides the *Query*, it is equivalent to the remote sensing imagery extracting standardized spatial structures and contour information from the map to prevent geometric distortion. Equations (2) and (3) express this computational process:

$$\hat{T}_{Map} = Softmax\left(\frac{Q_{RSI}{K_{Map}}^T}{\sqrt{d_k}}\right)V_{Map} \quad (2)$$

$$\hat{T}_{RSI} = Softmax\left(\frac{Q_{Map}{K_{RSI}}^T}{\sqrt{d_k}}\right)V_{RSI} \quad (3)$$

where Q, K, and V are linearly projected from the corresponding feature sequences T_{Map} and T_{RSI}, respectively; d_k is a scaling factor.

After interaction, the sequences pass through the Gated Fusion Network to output a fused sequence $T_f \in \mathbb{R}^{N\times D}$. Composed of two fully connected layers, the Gated Fusion Network achieves data source fusion by dynamically adjusting the contributions of the two paths.

3. Transformer Encoder Block.

The fused sequence T_f is fed into the stacked Transformer Encoder (Figure 2c) for deep semantic integration. The global receptive field of the MHA enables the effective capture of long-range spatial dependencies between settlement objects. The Latent Features $T_l \in \mathbb{R}^{N\times D}$ output by the Transformer Encoder Block are subsequently transmitted to the decoder.

3.2.2. Progressive Upsampling Decoder

The decoder is designed to map deep semantic features back to the spatial domain at the initial dimension. The latent features T_{latent} are first reshaped into a 3D feature map, followed by a stepwise restoration of image size via the progressive upsampling strategy.

Figure 2e illustrates the structure of the Upsampling Block and the transformations of feature shapes within it. Sub-pixel convolutional upsampling leverages rich channel information to amplify size, and effectively mitigates "Checkerboard Artifacts" without requiring additional training parameters [44]. Convolutional layers and the Spatial Attention mechanism are employed to rectify local blurring that may occur during the upsampling process. Furthermore, Skip Connection Features from shallow layers enhance the decoder's capability to reconstruct map details and spatial structures, and enable the model to focus on the discrepancies between the pre- and post-generalization maps.

3.3. Composite Loss Function Considering Settlement Generalization Characteristics

Settlement generalization is not merely a pixel classification task but also a process involving geometric transformation and morphological reconstruction. To address the specific characteristics of settlement generalization, a composite loss function $\mathcal{L}$ incorporating multiple constraints is constructed, defined as Equation (4). Here, α, β, γ and δ denote their respective weight coefficients.

$$\mathcal{L} = \alpha\mathcal{L}_{BCE} + \beta\mathcal{L}_{DICE} + \gamma\mathcal{L}_{Edge} + \delta\mathcal{L}_{SSIM} \quad (4)$$

1. Binary Cross Entropy Loss ($\mathcal{L}_{BCE}$).

Given that the target outputs are binary images, Binary Cross Entropy Loss is adopted as the basic loss function to measure pixel-level classification probability errors, defined as Equation (5):

$$\mathcal{L}_{BCE} = -\frac{1}{N}\sum_i^N [t_i \log(p_i) + (1 - t_i)\log(1 - p_i)] \tag{5}$$

where p_i and t_i denote the predicted value and target value of the i-th pixel, respectively, and N represents the total number of pixels.

2. DICE Loss ($\mathcal{L}_{DICE}$).

In certain map image tiles, the proportion of background pixels significantly exceeds that of foreground settlement pixels, leading to a substantial class imbalance issue. To address this, the DICE Loss is introduced to strengthen the model's attention to settlement pixels, defined as Equation (6):

$$\mathcal{L}_{DICE} = 1 - \frac{2\sum_i^N (p_i t_i)}{\sum_i^N p_i + \sum_i^N t_i} \tag{6}$$

3. Edge Enhancement Loss ($\mathcal{L}_{Edge}$).

The shape of generalized settlements is a critical factor influencing map legibility [45], and this shape is primarily defined by boundaries. Meanwhile, the incorporation of lower resolution remote sensing imagery may introduce boundary blurring. Therefore, an Edge Enhancement Loss is designed to enhance the model's constraint ability on cartographic specifications such as boundary straightness and orthogonality, defined as Equation (7). This loss term utilizes both Laplacian and Sobel operators to extract contour gradients and computes the $\mathcal{L}_1$ distance between the predicted and target gradients. Here, ∇_{Lap} and ∇_{Sobel} represent the corresponding gradient magnitudes. The Laplacian convolution kernel is defined as $\begin{bmatrix} 0 & -1 & 0 \\ -1 & 4 & -1 \\ 0 & -1 & 0 \end{bmatrix}$, while the horizontal and vertical kernels of the Sobel operator are $\begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{bmatrix}$ and $\begin{bmatrix} -1 & -2 & -1 \\ 0 & 0 & 0 \\ 1 & 2 & 1 \end{bmatrix}$, respectively.

$$\mathcal{L}_{Edge} = \frac{1}{N}\sum_i^N \left\| \nabla_{Lap}(p)_i - \nabla_{Lap}(t)_i \right\|_1 + \frac{1}{N}\sum_i^N \| \nabla_{Sobel}(p)_i - \nabla_{Sobel}(t)_i \|_1 \tag{7}$$

4. Structural Similarity Loss ($\mathcal{L}_{SSIM}$).

Preserving the spatial distribution patterns of building groups is a crucial metric for evaluating settlement generalization. This paper defines a loss function term based on the Structural Similarity Index Measure (SSIM) [46] to enforce fidelity regarding the macro spatial structure of the predicted images, defined as Equation (8):

$$\mathcal{L}_{SSIM} = 1 - \frac{(2\mu_p\mu_t + C_1)(2\sigma_{pt} + C_2)}{\left(\mu_p^2 + \mu_t^2 + C_1\right)\left(\sigma_p^2 + \sigma_t^2 + C_2\right)} \tag{8}$$

where μ_p and μ_p denote the pixel means of the predicted and target images, respectively; σ_p and σ_t represent the standard deviations; σ_{pt} is the covariance; and C_1 and C_2 are constants.

To investigate the effects of different constraints on the generalization results and avoid an overly empirical selection of loss weights, several loss function combinations were

predefined for comparative analysis. The experimental and analytical results are presented in Section 4.2.1.

4. Experiments and Analysis

The experiments aim to achieve two primary objectives: first, to investigate the matching relationship between remote sensing image resolution and the target scale of map generalization using the DSG-Trans model, providing a basis for image selection in multi-source data fusion; second, to validate the effectiveness of the dual-source fusion strategy and the DSG-Trans model in practical, cascaded multi-scale settlement generalization scenarios. Experiments were conducted under two scale transformation scenarios: 1:5k → 1:10k and 1:5k → 1:25k.

4.1. Experimental Settings

4.1.1. Dataset and Sample Construction

New York City, USA, was selected as the study area, as depicted in Figure 3. The study area governs five boroughs and is characterized by high urbanization, dense building groups, and distinct spatial distribution patterns. A multi-source dataset comprising source-scale raster maps, multi-resolution remote sensing imagery, and target-scale raster maps was constructed for the experiments.

- Source-Scale Raster Maps.

Figure 3. Study area and experimental data.

These were derived from the 3D-GloBFP building footprint dataset [47], which contains global building data collected in 2020 at a scale of 1:5k. Within the New York City building dataset, small buildings less than 100 m^2 and large buildings greater than 3000 m^2 account for about 27%, with diverse building morphologies and a wide area range, resulting in a certain imbalance in generalization complexity, posing a significant challenge to model performance. The vector settlement data was converted into raster maps with a resolution of 1 m/pixel. This resolution was determined by multiplying the standard map accuracy of 0.2 mm by the denominator of the 1:5k source scale.

- Remote Sensing Images.

Four open-source multi-resolution remote sensing images were adopted, all acquired in 2020 to ensure temporal alignment between the two data sources. The selected images include: Google Maps satellite imagery, integrating various commercial satellite and aerial data, with resolution reaching the sub-meter level; multispectral aerial orthophotos provided by the National Agriculture Imagery Program (NAIP), with a resolution of 0.6 m/pixel; Sentinel-2 satellite imagery, provided by the European Union's Coperni-

cus Programme, with a resolution of 10 m/pixel; and Landsat 8 satellite imagery, containing 30 m/pixel multispectral bands and a 15 m/pixel panchromatic band, obtaining 15 m/pixel multispectral data through pan-sharpening. To ensure data alignment, bilinear interpolation was employed to downsample some remote sensing images, ensuring integer dimensions for the image tiles. The resulting candidate multi-resolution imagery includes: Google-1 m/pixel, NAIP-2 m/pixel, NAIP-5 m/pixel, Sentinel-2–10 m/pixel, and Landsat 8–16 m/pixel.

- Target-scale Raster Maps.

The 1:10k and 1:25k settlement generalization maps were produced through a "software generalization + manual optimization" approach. Given that the visual perception changes induced by image downsampling primarily correspond to the simplification, aggregation, and selection operators, this paper focuses on these three operators. During automatic software generalization, the "Simplify Building" and "Aggregate Polygons" tools were utilized for preliminary generalization. Subsequently, several volunteers with cartographic domain knowledge conducted further manual optimization, which mainly involved correcting irrational adhesions between buildings and enhancing typical morphological features like boundary orthogonality and parallelism. The final generalization results are presented in Figure 3b.

This paper partitioned the complete raster maps into 320 × 320-pixel image tiles using the sliding window approach, a size that adequately covers the neighborhood relationships among settlements. A 50% overlap was maintained between tiles to mitigate artifacts during stitching and minimize the loss of contextual information. Correspondingly, the remote sensing imagery was partitioned into image tiles covering identical geographic extents, and the sizes of these tiles varied according to the resolution of the imagery. To avoid information leakage across different administrative regions, image tiling was conducted separately for each borough, and the data from different boroughs did not overlap spatially.

This process finally yielded a dataset containing 29,500 samples (Table 1). Data from four boroughs—Manhattan, Brooklyn, Queens, and The Bronx—were utilized for training, among which 15% of the samples were randomly allocated to the validation set for monitoring the training process and selecting hyperparameters. Data from Staten Island served as an independent test set to evaluate model performance and was excluded from both the training and validation processes.

Table 1. Number of settlements and sample pairs in the five boroughs of New York City at scales of 1:5k, 1:10k and 1:25k.

Dataset	Boroughs	Number of Settlements			Number of Sample Pairs
		1:5k	1:10k	1:25k	
Training Set	Manhattan	5452	4478	3631	6936
	Brooklyn	85,604	24,146	12,010	2375
	Queens	201,992	52,722	20,261	10,693
	The Bronx	42,629	16,854	7511	5234
Test Set	Staten Island	85,816	30,992	11,311	4262
Total	-	421,493	129,192	54,724	29,500

In addition, the probability maps generated by the model require post-processing to facilitate the production of standard vector maps. Initially, the image tiles were stitched into a complete image, employing a distance-weighted averaging method to fuse pixel values in overlapping regions, where pixels closer to the tile center were assigned higher weights. Subsequently, a threshold of 0.5 was applied to binarize the probability map into a raster

map, which then underwent vectorization and regularization. The purpose of regularization is to mitigate or eliminate jagged boundary artifacts resulting from vectorization, and rectify polygons into regular geometric shapes. Parameters allowing for arbitrary angles and a 2 m tolerance were adopted to avoid excessive modification of the model predictions during post-processing while preserving the original boundary orientations as much as possible.

4.1.2. Hardware Environment and Hyperparameter Settings

The model was implemented using Python 3.9 and PyTorch 2.1.2, with data-parallel training conducted on a workstation equipped with a 32 vCPU Intel(R) Xeon(R) Gold 6430 processor and dual NVIDIA GeForce RTX 4090 (24 GB) GPUs.

Following tests on various network architectures and hyperparameters, the following configuration was established: embedding dimension, D = 1024; patch size, P = 2; sequence length, N = 400; number of Transformer encoder layers, L = 12; number of attention heads, H = 8; the Adam optimizer, employed with an initial learning rate of 1×10^{-4}, gradually decaying to 1×10^{-6} via a cosine annealing strategy; a batch size of 16 and 100 epochs; and assigned values of 0.4, 0.2, 0.2, and 0.2, respectively, for the weights of $\mathcal{L}_{BCE}$, $\mathcal{L}_{DICE}$, $\mathcal{L}_{Edge}$ and $\mathcal{L}_{SSIM}$ in the composite loss function. These parameters were determined based on performance on the validation set, yielding a final model with approximately 0.2 billion parameters.

4.1.3. Evaluation Metrics

A quantitative evaluation framework was established encompassing both raster and vector dimensions. For raster results (i.e., binarized raster maps), Precision, Recall, F1-score, and pixel-level Intersection over Union (IoU) were employed to evaluate the fundamental classification performance of the model. Building upon this, Boundary IoU (BIoU) [48] was introduced to calculate the overlap of settlement boundary pixels, aiming to assess the shape fidelity of settlement contours, with its boundary width parameter set to 1. Meanwhile, the SSIM was utilized to quantify the preservation of spatial distribution patterns. For vector results (i.e., maps following vectorization and regularization), the evaluation focused on the generalization quality and geometric regularity of individual settlements. On the one hand, the object-level IoU between predicted and target features was calculated to analyze the distribution of settlement numbers across various IoU intervals. On the other hand, the rates of change in the number of parallel edges and right-angled corners pre- and post-generalization were computed to quantitatively analyze the model's effectiveness in preserving settlement parallelism and orthogonality.

It should be noted that map generalization is inherently a multi-solution problem. Provided that the results adhere to spatial cognitive principles and cartographic specifications, they are considered valid even if they exhibit certain deviations from the target data. Therefore, besides the above quantitative metrics, this paper also incorporates visual evaluations to qualitatively analyze the generalization results.

4.2. Multi-Scale Settlement Generalization Based on DSG-Trans Model

4.2.1. Comparison of Loss Weight Configurations

Before the formal scale-matching experiments, preliminary experiments were conducted in the 1:5k $\rightarrow$ 1:10k scale transformation scenario to examine the effects of different loss terms and determine a reasonable weight configuration. The experiments used remote sensing imagery with a spatial resolution of 2 m/pixel, and the number of training epochs was uniformly set to 50. As shown in Table 2, seven configurations were compared. G1 used only the Binary Cross Entropy loss, G2–G4 introduced an additional individual loss

term on this basis, and G5–G7 further compared the combined effects of the four loss terms under different weight configurations.

Table 2. Experimental results of different loss term combinations.

Configuration	Composite Loss Function	IoU	BIoU	SSIM
G1	$1.0\mathcal{L}_{BCE}$	0.6322	0.7823	0.8726
G2	$0.7\mathcal{L}_{BCE} + 0.3\mathcal{L}_{DICE}$	0.6538	0.6458	0.8568
G3	$0.7\mathcal{L}_{BCE} + 0.3\mathcal{L}_{Edge}$	0.8422	0.8337	0.9199
G4	$0.7\mathcal{L}_{BCE} + 0.3\mathcal{L}_{SSIM}$	0.7924	0.8268	0.9167
G5	$0.25\mathcal{L}_{BCE} + 0.25\mathcal{L}_{DICE} + 0.25\mathcal{L}_{Edge} + 0.25\mathcal{L}_{SSIM}$	0.8439	0.8506	0.9224
G6	$0.7\mathcal{L}_{BCE} + 0.1\mathcal{L}_{DICE} + 0.1\mathcal{L}_{Edge} + 0.1\mathcal{L}_{SSIM}$	0.8459	0.8559	0.9231
G7	$\mathbf{0.4\mathcal{L}_{BCE} + 0.2\mathcal{L}_{DICE} + 0.2\mathcal{L}_{Edge} + 0.2\mathcal{L}_{SSIM}}$	**0.8777**	**0.8707**	**0.9251**

Bold figures indicate the best results for a given metric.

The results show that introducing the DICE loss in G2 improved the IoU, but slightly reduced the BIoU and SSIM, indicating that it mainly benefited regional prediction while providing relatively limited constraints on boundary and overall structural fidelity. In contrast, G3 and G4 exhibited more consistent improvements. With the introduction of the Edge Enhancement loss, the IoU, BIoU, and SSIM increased by 21.00%, 5.14%, and 4.73%, respectively, compared with G1. The introduction of the structural similarity loss resulted in improvements of 16.02%, 4.45%, and 4.41%, respectively, demonstrating that these two loss terms contributed to preserving boundary shapes and maintaining spatial structural consistency, respectively. The results of G5–G7 further indicate that when all four loss terms were jointly applied, their contributions to model optimization were not identical, and simply assigning equal weights was insufficient to fully exploit the complementary effects of different constraints. Among the tested configurations, G7 achieved the highest IoU, BIoU, and SSIM. Therefore, the G7 weight configuration was adopted in the subsequent experiments.

4.2.2. Scale Matching Between Image Resolution and Target Scale

This experiment initially prepared multi-source, multi-resolution remote sensing imagery as candidate data, employing them to perform settlement generalization at specific target scales. The resulting maps were then compared from both quantitative and visual perspectives to identify the relatively better-performing resolutions for the 1:10k and 1:25k scales, which were subsequently regarded as the suitable resolutions matching the specific target scales.

Table 3 presents the quantitative evaluation of the results obtained using the candidate imagery. Here, "Only_Map" denotes a single-source input utilizing only raster maps, where the BCAFM is replaced by a single-layer Transformer. The results indicate that, compared to the single-source baseline, the introduction of dual-source data led to overall improvements in IoU, BIoU, and SSIM. Specifically, the 2 m/pixel and 5 m/pixel resolutions achieved relatively better overall performance for the 1:10k and 1:25k scales, respectively, with IoU values exceeding the "Only_Map" results by 3.01% and 3.96%. This suggests that these specific resolutions can provide effective scale guidance for the corresponding generalization process. Additionally, as the spatial resolution of the remote sensing imagery decreases, the model's FLOPs show an overall downward trend. Although the 1 m/pixel resolution achieved suboptimal metrics, its training time was more than double that of the better-performing schemes. Thus, selecting lower-resolution remote sensing imagery can reduce computational costs while maintaining satisfactory generalization quality.

Table 3. Quantitative evaluation results using multi-resolution candidate imagery at target scales of 1:10k and 1:25k.

Target Scales	Candidate Remote Sensing Images	IoU	BIoU	SSIM	Training Time (Hours)	FLOPs (G)
1:10k	Google-1 m/pixel	0.8996	0.8940	0.9301	30.69	248.71
	NAIP-2 m/pixel	**0.9007**	**0.8948**	**0.9305**	14.75	182.75
	NAIP-5 m/pixel	0.8986	0.8931	0.9293	12.15	162.28
	Sentinel-2–10 m/pixel	0.8982	0.8928	0.9294	12.10	161.64
	Landsat 8–16 m/pixel	0.8989	0.8935	0.9295	11.85	161.11
	Only_Map	0.8706	0.8686	0.9282	11.12	139.78
1:25k	Google-1 m/pixel	0.8085	0.7395	0.8902	30.41	248.71
	NAIP-2 m/pixel	0.8024	0.7314	0.8872	15.08	182.75
	NAIP-5 m/pixel	**0.8124**	**0.7415**	**0.8903**	12.25	162.28
	Sentinel-2–10 m/pixel	0.8053	0.7348	0.8886	12.10	161.64
	Landsat 8–16 m/pixel	0.8076	0.7373	0.8899	11.81	161.11
	Only_Map	0.7728	0.7052	0.8865	10.47	139.78

Bold figures indicate the best results for a given metric, and underlined figures indicate the second-best results.

To further examine the statistical significance of the above results, the test area was divided into 47 non-overlapping 2 km × 2 km spatial units, and IoU, BIoU, and SSIM were calculated separately for each unit. On this basis, the 2 m/pixel and the 5 m/pixel resolutions were used as the reference schemes and compared with the other input schemes using the paired Wilcoxon signed-rank test. The results are presented in Table 4. For the 1:10k task, the 2 m/pixel exhibited statistically significant differences ($p < 0.05$) from the 5, 10, and 16 m/pixel and the "Only_Map" for most evaluation metrics. Compared with the 1 m/pixel, significant differences were observed in IoU and BIoU, whereas the difference in SSIM was not statistically significant. For the 1:25k task, the 5 m/pixel showed significant differences from the 2 m/pixel, 10 m/pixel, and "Only_Map" across all three metrics, whereas some metrics did not reach statistical significance when compared with the 1 m/pixel and 16 m/pixel. Overall, the statistical analysis further indicates that the 2 m/pixel and 5 m/pixel provide relatively stable overall performance at their corresponding target scales, while the differences between some adjacent image resolutions remain limited.

Table 4. Statistical significance of performance differences among different input schemes.

Target Scales	Comparison	IoU *p*-Value	BIoU *p*-Value	SSIM *p*-Value
1:10k	2 m/pixel vs. 1 m/pixel	**0.0085**	**0.0121**	0.3102
	2 m/pixel vs. 5 m/pixel	**<0.001**	**<0.001**	**<0.001**
	2 m/pixel vs. 10 m/pixel	**<0.001**	**<0.001**	**<0.001**
	2 m/pixel vs. 16 m/pixel	**<0.001**	**0.0036**	**<0.001**
	Only_Map	**<0.001**	**<0.001**	**<0.001**
1:25k	2 m/pixel vs. 1 m/pixel	**0.0287**	0.0556	0.5118
	2 m/pixel vs.5 m/pixel	**<0.001**	**<0.001**	**<0.001**
	2 m/pixel vs. 10 m/pixel	**<0.001**	**<0.001**	**<0.001**
	2 m/pixel vs. 16 m/pixel	**<0.001**	**0.0018**	0.2905
	Only_Map	**<0.001**	**<0.001**	**<0.001**

The *p*-values were obtained using the paired Wilcoxon signed-rank test. Bold values indicate statistically significant differences at $p < 0.05$.

Figure 4 provides a more intuitive visual comparison of the generalization results obtained using the candidate imagery. At the 1:10k target scale, the 2 m/pixel results

exhibit superior visual performance. For instance, Region A demonstrates that two settlements with complex contours should be simplified into rectangular shapes; the 2 m/pixel result approximates this objective most closely, whereas other results display incomplete simplification. Similarly, in Region D, only the 2 m/pixel result correctly aggregated three settlements and generated regular geometric shapes. Furthermore, results from other resolutions exhibited artifacts such as small fragments (e.g., B, E) and boundary erosion (e.g., C). At the 1:25k target scale, the 5 m/pixel results are more favorable, while other results displayed unsatisfactory generalization decisions, such as the failure to eliminate settlements in Regions I and J. However, the advantages of the 5 m/pixel results are more pronounced in the preservation of settlement boundary parallelism and orthogonality. In contrast, results from other resolutions produced a noticeably higher number of settlements with irregular shapes, such as Regions H, I and J. This confirms the positive contribution of appropriate scale information to the adherence to cartographic rules.

Figure 4. Settlement generalization results using multi-resolution candidate imagery at target scales of 1:10k and 1:25k. Each region covers an identical extent of approximately 280 m × 240 m. The coordinates indicate the longitude and latitude of the center point for each region; this convention applies to all subsequent result figures in this paper. A–I indicate representative example areas used for illustration and discussion in the text.

4.2.3. Analysis of Training Convergence

Based on the above matching results, this paper analyzes in detail the performance of using 2 m/pixel and 5 m/pixel imagery in practical multi-scale settlement general-

ization tasks from three perspectives: training convergence, visual performance, and quantitative evaluation.

Figure 5 illustrates the training IoU and loss curves across the two scale transformation scenarios. The IoU for both training and validation sets exhibits a synchronized upward trend, while the loss values show a synchronized decline, indicating effective model convergence within the limited training epochs. Due to the larger scale span in the 1:5k → 1:25k scenario, the increased complexity of the generalization process results in slightly inferior convergence compared to the 1:5k → 1:10k scenario. Ultimately, the training and validation IoU at the 1:10k target scale reached 94.41% and 94.14%, respectively, while those at the 1:25k target scale were 90.22% and 89.56%.

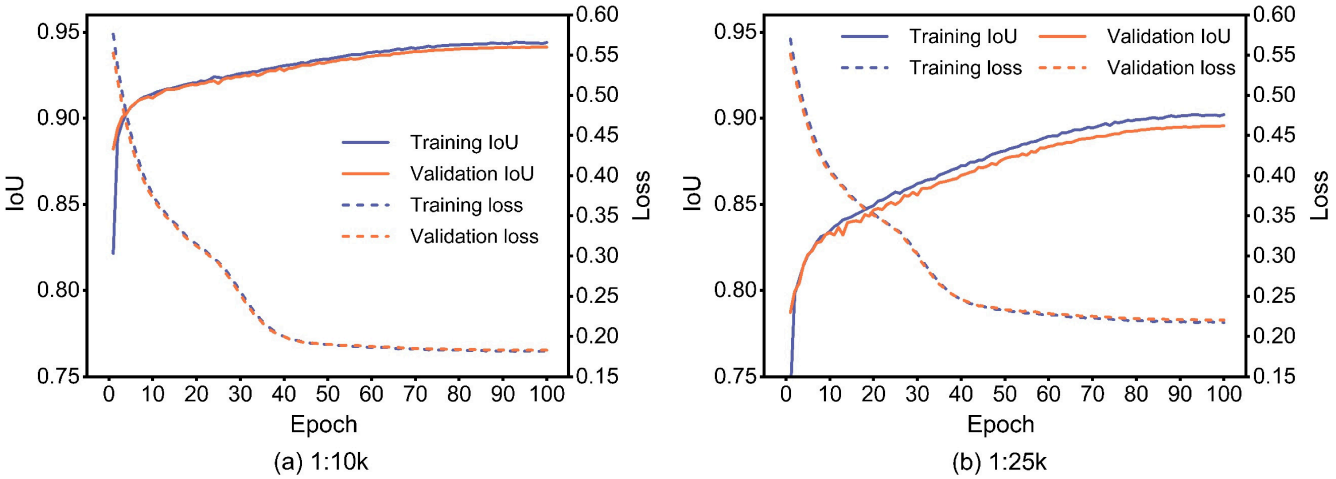

Figure 5. IoU and loss curves during the model training process.

4.2.4. Visual Effect Evaluation

Figure 6 presents zoomed-in views of randomly selected regions from the test set to illustrate the efficacy of multi-scale settlement generalization using the proposed method. It can be seen that the model executed different generalization operators in various situations: settlements with complex contour details are simplified into rectangles (e.g., A → C, E → G); similar settlements close in distance are aggregated (e.g., B → D, E → H); insignificant settlements are eliminated (e.g., F → I, J → M); and the predicted results also demonstrate distinct cascading generalization decisions changing with scale (e.g., K → L → N, O → P → Q). Correspondingly, within the remote sensing imagery at the same locations, these settlements appear with blurred boundaries or are even indistinguishable.

Furthermore, settlement boundaries in most regions exhibit evident straightness and a tendency toward right-angled corners, verifying that the model architecture and loss function constraints effectively preserve parallelism and orthogonality. However, corners in the 1:25k results are relatively smoother, indicating that larger scale spans pose greater challenges to preserving settlement boundary morphology.

The "Vector Map" shown in Figure 6 is the result of vectorizing and regularizing the model prediction (Prediction). After these automated post-processing steps, the pixelated jagged edges of the buildings are significantly reduced, and the resulting vector features exhibit clearer boundaries and more regular shapes. This provides a basis for subsequent map production and standardization.

Figure 6. Multi-scale settlement generalization results obtained using the proposed method. The figure includes the source-scale raster maps (Source), multi-resolution remote sensing imagery (RSI), model predictions (Prediction), target maps (Target), and the vectorized and regularized model predictions (Vector Map). The images presented in the "Vector Map" column represent results without any manual post-processing. Each region covers an identical extent of approximately 260 m × 220 m. A–Q indicate representative example areas used for illustration and discussion in the text.

To demonstrate the performance of the predicted results at actual sizes, Figure 7 presents the generalization results for two larger regions. It is evident that the maps at different scales adapt well to their respective scale requirements, exhibiting rationality consistent with human visual perception regarding shape simplification, spatial distribution, and overall legibility. It is worth noting that although the 1:25k results are slightly deficient in local details compared to the 1:10k results, the boundary distortions and artifacts observed in the zoomed-in views become indiscernible at the actual map scale.

Figure 7. Multi-scale settlement generalization results corresponding to actual scale relationships. Each region covers an identical extent of approximately 1.7 km × 1 km.

4.2.5. Quantitative Evaluation

Table 5 presents a quantitative evaluation of the settlement generalization results using 2 m/pixel and 5 m/pixel imagery, conducted in raster format. Judging from the comparison between model predictions and target maps ("Prediction vs. Target"), both the 1:10k and 1:25k results exhibited superior F1-scores (exceeding 94% and 89%, respectively) and high IoU values (90.07% and 81.24%, respectively), demonstrating the model's capability to accurately execute generalization decisions across different scale spans. Meanwhile, the favorable performance regarding BIoU (89.48% and 74.15%) and SSIM (93.05% and 89.03%) confirms the model's capability to preserve boundary details while maintaining the macro-spatial distribution patterns of building groups.

Table 5. Quantitative evaluation of settlement generalization results in raster format.

Target Scales	Comparative Data	Precision	Recall	F1-Score	IoU	BIoU	SSIM
1:10k	Source vs. Target	0.9192	0.8216	0.8666	0.7684	0.8172	0.8780
	Prediction vs. Target	0.9478	0.9472	0.9475	0.9007	0.8948	0.9305
1:25k	Source vs. Target	0.8326	0.6638	0.7359	0.5872	0.6015	0.8146
	Prediction vs. Target	0.9007	0.8913	0.8957	0.8124	0.7415	0.8903

Given that generalization complexity significantly influences metric values, relying solely on the similarity between predictions and targets ("Prediction vs. Target") is insufficient. Instead, the metric gains achieved relative to the source data may better reflect the model's performance in complex generalization tasks. Specifically, this gain refers to the improvement of "Prediction vs. Target" over "Source vs. Target". Table 5 shows that at the 1:10k target scale, the "Source vs. Target" IoU is relatively low (76.84%), indicating the high complexity and uncertainty associated with the map generalization operations involved. In terms of metric gains, IoU, BIoU, and SSIM increased by 13.23%, 7.76%, and 5.25%, respectively, demonstrating effective improvement by the model. At the 1:25k target scale, the "Source vs. Target" IoU is only 58.72%, attributed to the significantly higher generalization complexity resulting from the larger scale span. Consequently, the absolute metrics for the 1:25k results are slightly lower than those for the 1:10k results. However, the obvious gains in IoU, BIoU, and SSIM (22.52%, 14.00%, and 7.57%, respectively) indicate that the model retains the capability to effectively address complex map generalization problems.

Following vectorization, regularization, and the removal of noise and fragments smaller than 10 pixels, the model predicted 31,417 and 11,184 settlements for the 1:10k and 1:25k scales, respectively, corresponding to relative errors of only 1.37% and 1.12% compared to the targets.

Table 6 presents the quantitative distribution of predicted settlements based on object-level IoU at the individual level. Here, the object-level IoU means the degree of overlap between predicted and target settlements in vector format. Given that map generalization is inherently a multi-solution problem where predictions cannot perfectly align with targets, we consider $IoU_{obj} > 0.75$ to be a threshold indicating high generalization quality. Table 6 reveals that the proportions of settlements with $IoU_{obj} > 0.75$ in the 1:10k and 1:25k results are 87.12% and 77.46%, respectively. These results demonstrate that the vast majority of predicted settlements maintain high spatial consistency with the target objects in terms of both positional accuracy and shape fidelity.

Table 6. Distribution of object-level IoU for generalization results at the individual settlement level.

Target Scale	IoU Interval	Count	Percentage (%)
1:10k	0.75—1.00	27,372	87.12
	0.50—0.75	1970	6.27
	0.25—0.50	1118	3.56
	0.00—0.25	957	3.05
1:25k	0.75—1.00	8663	77.46
	0.50—0.75	1462	13.07
	0.25—0.50	533	4.77
	0.00—0.25	526	4.70

"Count" and "Percentage" indicate the number and percentage of predicted settlements whose IoU values with the corresponding target settlements fall within specific intervals.

Table 7 presents statistics regarding the percentage of right-angled corners to total internal angles, as well as the percentage of parallel edges to the total number of edges, pre- and post-generalization. Both indicators are calculated from the vectorized settlement polygons. Specifically, the interior angles at polygon vertices and the directional relationships between polygon edges are analyzed to determine the proportions of right-angled corners and parallel edges, respectively, with an angular tolerance of 0.5°. It is evident that the proportions of right-angled corners and parallel edges in both the 1:10k and 1:25k results increase noticeably, which is consistent with the general principle of regularizing building morphology during map generalization, with the improvement being more pronounced at

1:10k. It should be noted that the 2 m tolerance with arbitrary angles adopted in this study represents a relatively conservative setting, intended to avoid excessive modification of the model predictions during post-processing. For smaller target scales such as 1:25k, a larger ground tolerance could theoretically be adopted. The relatively limited improvements in the proportions of right-angled corners and parallel edges at 1:25k are more likely related to the larger scale transition and the increased complexity of the generalization operations.

Table 7. Maintenance of parallelism and orthogonality in settlement generalization results.

Target Scales	Percentage of Right-Angled Corners (%)			Percentage of Parallel Edges (%)		
	Pre-Generalization	Post-Generalization	Change	Pre-Generalization	Post-Generalization	Change
1:10k	81.68	96.57	+14.88	87.98	98.54	+10.56
1:25k	81.68	83.12	+1.43	87.98	93.19	+5.21

In summary, the 2 m/pixel and 5 m/pixel remote sensing imagery can serve as the optimal scale guidance data for generalization tasks at target scales of 1:10k and 1:25k, respectively. Based on this scale correspondence, the DSG-Trans model, employing a dual-source fusion strategy, is capable of generating high-quality generalization results, thereby fully validating the effectiveness and reliability of the proposed method.

4.3. Comparison with Other Methods

In image generation tasks, considering the demonstrated advantages of ResU-Net and GAN models, ResU-Net [49] and pix2pix [50] were selected as comparative models for the proposed DSG-Trans model. Simultaneously, control experiments with single-source and dual-source inputs were designed for all three models to validate the effectiveness of the dual-source fusion strategy. Thus, we obtained six methods across different models and input data sources. For the ResU-Net model, the initial learning rate was set to 1×10^{-3}, decaying by a factor of 0.1 every 20 epochs, with a batch size of 16 and a total of 50 epochs. For the pix2pix model, the learning rate, batch size, and total epochs were set to 2×10^{-4}, 16, and 100, respectively. Model architectures and hyperparameters were determined by referencing the original publications and through testing on the current dataset.

Table 8 presents the experimental results for different methods on the test set, where "SS" and "DS" denote Single-Source and Dual-Source inputs, respectively. It is evident that the proposed method ("DSG-Trans_DS") achieves the best metrics at both the 1:10k and 1:25k target scales. From the comparison of different input data sources, except for the pix2pix model at the 1:10k scale, all results indicate that dual-source inputs outperform single-source inputs. The pix2pix model may not be effective at fusing information from two types of data. In terms of model comparison, the overall performance of the DSG-Trans model is significantly superior to that of the other comparative models.

Figure 8 provides a visual comparison of the settlement generalization results obtained by different methods. It is evident that the proposed method ("DSG-Trans_DS") yields superior results compared to other methods (e.g., A, B, C). The results produced by the ResU-Net model exhibit incomplete generalization and fail to maintain the straightness of settlement boundaries. Additionally, the generalization results derived without the use of remote sensing imagery are noticeably inferior. Although the pix2pix model generates relatively sharp settlement boundaries, the results still suffer from significant boundary distortion and erosion. It is worth noting that while the results in Table 7 do not conclusively demonstrate the superiority of dual-source inputs for the pix2pix model, Figure 8 reveals that the dual-source results are visually superior.

Table 8. Quantitative evaluation results using different models and data source inputs.

Target Scale	Methods	IoU	BIoU	SSIM
1:10k	ResU-Net_DS	0.8344	0.8285	0.8967
	ResU-Net_SS	0.7997	0.8142	0.8855
	pix2pix_DS	0.8171	0.7951	0.8900
	pix2pix_SS	0.8366	0.8278	0.8948
	DSG-Trans_DS	**0.9007**	**0.8948**	**0.9305**
	DSG-Trans_SS	0.8706	0.8686	0.9282
1:25k	ResU-Net_DS	0.6565	0.6058	0.8399
	ResU-Net_SS	0.6324	0.6003	0.8318
	pix2pix_DS	0.6829	0.5666	0.8387
	pix2pix_SS	0.6483	0.5312	0.8280
	DSG-Trans_DS	**0.8124**	**0.7415**	**0.8903**
	DSG-Trans_SS	0.7728	0.7052	0.8865

Bold figures indicate the best results for dual-source inputs, while underlined figures indicate the best results for single-source inputs.

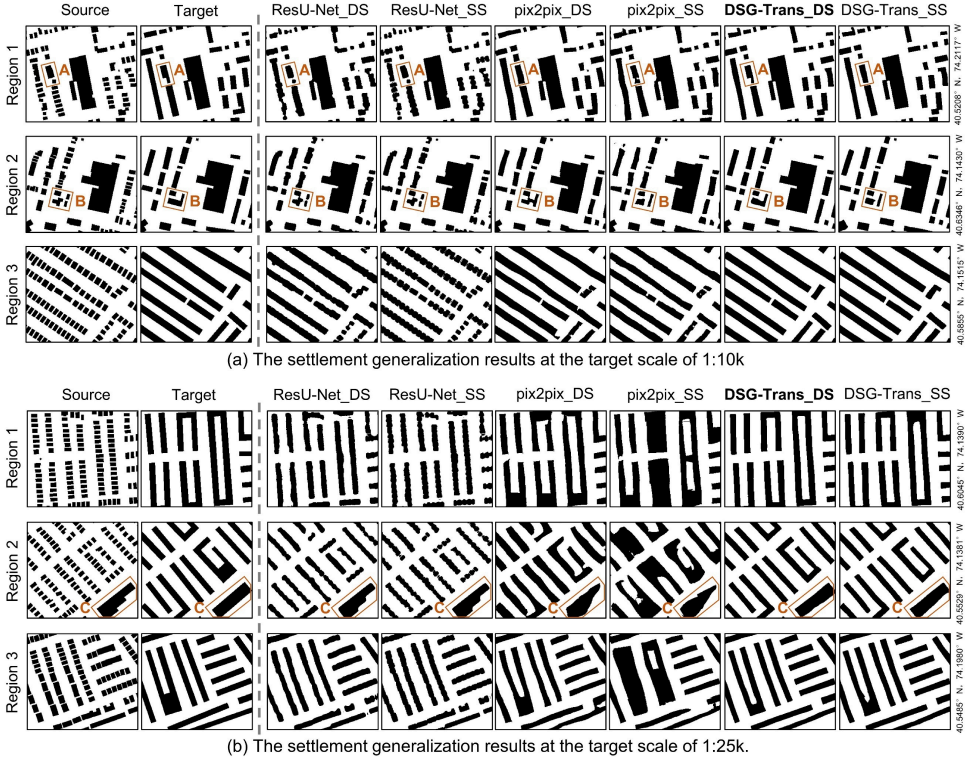

Figure 8. Settlement generalization results using different models and data source inputs at target scales of 1:10k and 1:25k. Each region covers an identical extent of approximately 280 m × 240 m. A–C indicate representative example areas used for illustration and discussion in the text.

These findings indicate that settlement generalization methods integrating maps and remote sensing imagery offer certain advantages. Furthermore, the proposed DSG-Trans model effectively bridges the information gap between the two data sources, yielding generalization results that surpass those of the comparative models.

4.4. Generalizability Analysis of the Model

To evaluate the generalization capability of the proposed model across datasets with varying morphologies and spatial distribution patterns, the model trained on the New York City dataset was applied to Downtown Los Angeles (DTLA), the core urban area of Los Angeles, California, USA.

The characteristics of building groups in DTLA differ significantly from those in New York City, featuring more complex building contours and a greater diversity of block patterns. The raster maps and remote sensing imagery utilized in this experiment were derived from the same data sources for this region and underwent identical data processing procedures. At the source scale, the DTLA dataset contains 27,709 buildings, yielding 1689 samples after image splitting. Due to the high cost of cartographic production and the difficulty of ensuring fully consistent production standards with the existing target data, corresponding 1:10k and 1:25k target generalization maps are currently unavailable for the DTLA area. This experiment primarily evaluates the generalization results from the perspective of visual performance.

Figure 9 presents the settlement generalization results in DTLA, where the numbers of settlements at the 1:10k and 1:25k target scales are 8266 and 2655, respectively. The results indicate that the model remains capable of executing reasonable generalization decisions—such as selection, simplification, and aggregation—in most regions, while preserving the original spatial distribution patterns of building groups to a significant extent (e.g., A, B, C, D). This suggests that the model possesses a certain degree of generalizability, enabling the transfer of acquired cartographic knowledge to regions with significantly different settlement characteristics. In the 1:10k results, the model excelled in maintaining the parallelism and orthogonality of settlement contours, but the 1:25k results exhibited some boundary irregularities and rounded corners.

Figure 9. Settlement generalization results in DTLA obtained using the proposed method. Region 1 and Region 2 cover an identical extent of approximately 1.7 km × 1 km. Sub-regions A, B, C, and D represent zoomed-in views for detailed observation.

Overall, the model demonstrates a certain degree of generalization capability in cross-regional scenarios. However, the overall performance in DTLA is inferior to that on the New York City dataset, indicating that the model's generalizability is constrained by specific settlement morphological characteristics and spatial patterns.

5. Discussion

5.1. Effectiveness Analysis of the Proposed Method

The effectiveness of the proposed method can be analyzed from both data and model perspectives. At the data level, the dual-source data fusion strategy breaks the information silos inherent in a single data source. Building upon single-source map generalization, the scale-matched remote sensing images introduce appropriate implicit scale and context information. On the one hand, remote sensing images are utilized to guide the generalization degree of map features; on the other hand, map constraints are employed to prevent morphological distortions and decision errors caused by low-resolution images. Furthermore, the raster-based solution offers distinct advantages in coordinating multiple generalization operators and can integrate remote sensing images more naturally.

At the model level, the DSG-Trans demonstrates outstanding map generalization performance. The self-attention mechanism of the Transformer overcomes the limitations of CNNs' local receptive fields, effectively modeling long-range dependencies between settlements within a unified feature space. The bidirectional cross-attention mechanism facilitates the fusion of map and remote sensing data in a high-dimensional space, achieving feature complementarity through bidirectional information flow. The skip connection strategy allows the model to focus more on the differences in settlement features before and after generalization. Moreover, the composite loss function tailored to the characteristics of settlement generalization exhibits advantages in preserving cartographic specifications.

5.2. Limitations and Future Work

Despite the positive progress achieved in this study in multi-scale settlement generalization, several limitations remain, which can be further explored with more innovative approaches in future studies. First, while generalizing in the raster space facilitates end-to-end modeling and multi-source data fusion, it inevitably introduces noise and jagged boundary artifacts, making it difficult to match the geometric accuracy of vector generalization. Future work could explore collaborative modeling approaches that fuse vector maps with remote sensing images, investigating information interaction mechanisms between vector coordinate sequences and raster images to establish an end-to-end multimodal model for better map generalization results. Second, the current model requires a specific resolution ratio between the input raster map and remote sensing images; such strict preprocessing limits model flexibility to some extent and may cause information loss during downsampling. Future research could focus on feature alignment mechanisms to enhance the compatibility and robustness of the model toward heterogeneous data sources. Third, remote sensing images are susceptible to complex environmental factors like illumination and occlusion; although this paper utilizes the CBAM for feature calibration, uncertainty risks remain, as the model might misinterpret image information and thereby affect generalization decisions. Subsequent studies could introduce more robust cross-modal representation learning techniques to mitigate the impact of image misinterpretation on the generalization results. Fourth, this study currently selects only ResU-Net and pix2pix as comparative models. Future work will further introduce more recent deep learning methods for map generalization under unified data formats and experimental conditions for systematic comparison. Finally, since experiments in this study were limited to two target scales (1:10k and 1:25k) using five candidate image resolutions, it is necessary to construct a sample

library covering more cartographic scales in the future to further verify the applicability and generalization capability of the proposed method.

6. Conclusions

This paper presents a settlement generalization method that integrates maps and remote sensing imagery, aiming to address the challenges of limited automation and reliance on manual expertise. Leveraging the correspondence between map scale and image resolution, the DSG-Trans model was developed to achieve deep synergy between the explicit geometric features of maps and the implicit scale information of remote sensing imagery. Furthermore, a composite loss function was incorporated to effectively preserve cartographic specifications, such as boundary parallelism and orthogonality. Experimental results demonstrate that, at target scales of 1:10k and 1:25k, the proposed method efficiently executes generalization operators while maintaining these strict geometric constraints alongside spatial structural consistency. This study validates the potential of remote sensing imagery to serve as scale guidance in map generalization, offering a novel theoretical perspective and technical pathway for achieving intelligent map production that integrates data-driven and knowledge-guided approaches.

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References

1. Wang, J. Cartography in the age of spatio-temporal big data. *Acta Geod. Cartogr. Sin.* **2017**, *46*, 1226–1237. [CrossRef]
2. Yu, W.; Zhang, Y.; Ai, T.; Guan, Q.; Chen, Z.; Li, H. Road network generalization considering traffic flow patterns. *Int. J. Geogr. Inf. Sci.* **2019**, *34*, 119–149. [CrossRef]
3. Gonzales, J.J. Building-Level Comparison of Microsoft and Google Open Building Footprints Datasets. In *12th International Conference on Geographic Information Science (GIScience 2023)*; Schloss Dagstuhl–Leibniz-Zentrum für Informatik: Wadern, Germany, 2023; pp. 1–35. [CrossRef]
4. Vaswani, A.; Shazeer, N.; Parmar, N.; Uszkoreit, J.; Jones, L.; Gomez, A.N.; Kaiser, L.; Polosukhin, I. Attention is all you need. *arXiv* **2017**, arXiv:1706.03762. [CrossRef]
5. Töpfer, F.; Pillewizer, W. The principles of selection. *Cartogr. J.* **1966**, *3*, 10–16. [CrossRef]
6. Regnauld, N.; Edwardes, A.; Barrault, M. Strategies in building generalization: Modelling the sequence, constraining the choice. In Proceedings of the ICA Workshop on Progress in Automated Map Generalization, Ottawa, ON, Canada, 12–14 August 1999; pp. 372–378.
7. Sester, M. Generalization based on least squares adjustment. *Int. Arch. Photogramm. Remote Sens.* **2000**, *33*, 931–938.
8. Yan, X.; Ai, T.; Zhang, X. Template matching and simplification method for building features based on shape cognition. *ISPRS Int. J. Geo-Inf.* **2017**, *6*, 250. [CrossRef]

9. Damen, J.; van Kreveld, M.; Spaan, B. High quality building generalization by extending the morphological operators. In Proceedings of the 11th ICA Workshop on Generalization and Multiple Representation, Montpellier, France, 20–21 June 2008; pp. 1–12.
10. Cámara, M.; López, F. Mathematical morphology applied to raster generalization of urban city block maps. *Cartographica* **2000**, *37*, 33–48. [CrossRef]
11. Schylberg, L. Cartographic amalgamation of area objects. *Int. Arch. Photogramm. Remote Sens.* **1993**, *29*, 135.
12. Su, B.; Li, Z.; Lodwick, G.; Muller, J. Algebraic models for the aggregation of area features based upon morphological operators. *Int. J. Geogr. Inf. Sci.* **1997**, *11*, 233–246. [CrossRef]
13. Jones, C.B.; Bundy, G.L.; Ware, M.J. Map generalization with a triangulated data structure. *Cartogr. Geogr. Inf. Syst.* **1995**, *22*, 317–331.
14. Li, Z.; Yan, H.; Ai, T.; Chen, J. Automated building generalization based on urban morphology and Gestalt theory. *Int. J. Geogr. Inf. Sci.* **2004**, *18*, 513–534. [CrossRef]
15. Ai, T.; Guo, R.; Chen, X. Simplification and aggregation of polygon object supported by delaunay triangulation structure. *J. Image Graph.* **2001**, *6*, 93–99.
16. Ware, J.M.; Jones, C.B.; Thomas, N. Automated map generalization with multiple operators: A simulated annealing approach. *Int. J. Geogr. Inf. Sci.* **2003**, *17*, 743–769. [CrossRef]
17. Wilson, I.D.; Ware, J.M.; Ware, J.A. Reducing graphic conflict in scale reduced maps using a genetic algorithm. In Proceedings of the 5th Workshop on Progress in Automated Map Generalization (ICA Commission on map generalization), Paris, France, 28–30 April 2003.
18. Ruas, A. A method for building displacement in automated map generalization. *Int. J. Geogr. Inf. Sci.* **1998**, *12*, 789–803. [CrossRef]
19. Ai, T. A displacement of building cluster based on field analysis. *Acta Geod. Cartogr. Sin.* **2004**, *33*, 89–94. [CrossRef]
20. Yan, X.; Yang, M.; Ai, T. Deep learning in automatic map generalization: Achievements and challenges. *Geo-Spat. Inf. Sci.* **2025**, *28*, 2905–2926. [CrossRef]
21. Touya, G.; Zhang, X.; Lokhat, I. Is deep learning the new agent for map generalization? *Int. J. Cartogr.* **2019**, *5*, 142–157. [CrossRef]
22. Wu, F.; Du, J.; Qian, H.; Zhai, R. Overview of research progress and reflections in intelligent map generalization. *Geomat. Inf. Sci. Wuhan Univ.* **2022**, *47*, 1675–1687. [CrossRef]
23. Liu, C.; Hu, Y.; Li, Z.; Xu, J.; Han, Z.; Guo, J. TriangleConv: A deep point convolutional network for recognizing building shapes in map space. *ISPRS Int. J. Geo-Inf.* **2021**, *10*, 687. [CrossRef]
24. Hu, Y.; Liu, C.; Li, Z.; Xu, J.; Han, Z.; Guo, J. Few-shot building footprint shape classification with relation network. *ISPRS Int. J. Geo-Inf.* **2022**, *11*, 311. [CrossRef]
25. Yan, X.; Ai, T.; Yang, M.; Tong, X. Graph convolutional autoencoder model for the shape coding and cognition of buildings in maps. *Int. J. Geogr. Inf. Sci.* **2021**, *35*, 490–512. [CrossRef]
26. Zhao, R.; Ai, T.; Yu, W.; He, Y.; Shen, Y. Recognition of building group patterns using graph convolutional network. *Cartogr. Geogr. Inf. Sci.* **2020**, *47*, 400–417. [CrossRef]
27. Fu, C.; Zhou, Z.; Xin, Y.; Weibel, R. Reasoning cartographic knowledge in deep learning-based map generalization with explainable AI. *Int. J. Geogr. Inf. Sci.* **2024**, *38*, 2061–2182. [CrossRef] [PubMed]
28. Feng, Y.; Thiemann, F.; Sester, M. Learning Cartographic Building Generalization with Deep Convolutional Neural Networks. *ISPRS Int. J. Geo-Inf.* **2019**, *8*, 258. [CrossRef]
29. Sester, M.; Feng, Y.; Thiemann, F. Building generalization using deep learning. *Int. Arch. Photogramm. Remote Sens. Spat. Inf. Sci.* **2018**, *XLII-4*, 565–572. [CrossRef]
30. Zhou, Z.; Fu, C.; Weibel, R. SpaGAN: A spatially-aware generative adversarial network for building generalization in image maps. *Int. J. Appl. Earth Obs. Geoinf.* **2024**, *135*, 104236. [CrossRef]
31. Du, J.; Wu, F.; Xing, R.; Li, C.; Li, J. Trial and comparison of some encoder-decoder based deep learning models for automated generalization of buildings. *Geomat. Inf. Sci. Wuhan Univ.* **2022**, *47*, 1052–1062. [CrossRef]
32. An, X.; Zhu, Y.; Yan, X. A building selection method supported by graph convolutional network. *Acta Geod. Cartogr. Sin.* **2023**, *52*, 1574–1583.
33. Xu, L.; Kong, B.; Xiao, T.; Yu, H.; Ai, T. Building aggregation method based on graph convolutional neural network. *Geospat. Inf.* **2024**, *22*, 10–14. [CrossRef]
34. Zhou, Z.; Fu, C.; Weibel, R. Move and remove: Multi-task learning for building simplification in vector maps with a graph convolutional neural network. *ISPRS J. Photogramm. Remote Sens.* **2023**, *202*, 205–218. [CrossRef]
35. Yan, X.; Yang, M. A deep learning approach for polyline and building simplification based on graph autoencoder with flexible constraints. *Cartogr. Geogr. Inf. Sci.* **2024**, *51*, 79–96. [CrossRef]
36. Feng, Y.; Qian, N.; Fu, C.; Zhou, Z.; Sester, M. PolygonTranslator—Learning to simplify building footprints from one scale to another. *Abstr. ICA* **2023**, *6*, 60. [CrossRef]

37. Cui, L.; Xu, J.; Jiang, L.; Qian, H. End-to-end vector simplification for building contours via a sequence generation model. *ISPRS Int. J. Geo-Inf.* **2025**, *14*, 124. [CrossRef]
38. Liu, P.C.; Cheng, X.Q.; Xiao, T.Y.; Yang, M.; Ai, T.H. A Transformer model for building polygon simplification in map generalization. *Acta Geod. Cartogr. Sin.* **2026**, *55*, 124–137.
39. Hu, Y.; Wang, Z.; Huang, Z.; Liu, Y. PolyBuilding: Polygon transformer for building extraction. *ISPRS J. Photogramm. Remote Sens.* **2023**, *199*, 15–27. [CrossRef]
40. Zhang, M.; Liu, Q.; Wang, Y. HiT: Building mapping with hierarchical Transformers. *IEEE Trans. Geosci. Remote Sens.* **2024**, *62*, 5606316. [CrossRef]
41. Woo, S.; Park, J.; Lee, J.; Kweon, I.S. CBAM: Convolutional block attention module. In *Computer Vision—ECCV 2018, Proceedings of the 15th European Conference, Munich, Germany, 8–14 September 2018*; Springer: Cham, Switzerland, 2018; pp. 3–19. [CrossRef]
42. Lu, J.; Batra, D.; Parikh, D.; Lee, S. ViLBERT: Pretraining task-agnostic visiolinguistic representations for vision-and-language tasks. *arXiv* **2019**, arXiv:1908.02265. [CrossRef]
43. Shen, J.; Chen, Y.; Liu, Y.; Zuo, X.; Fan, H.; Yang, W. ICAFusion: Iterative cross-attention guided feature fusion for multispectral object detection. *Pattern Recognit.* **2023**, *145*, 109913. [CrossRef]
44. Shi, W.; Caballero, J.; Huszár, F.; Totz, J.; Aitken, A.P.; Bishop, R.; Rueckert, D.; Wang, Z. Real-time single image and video super-resolution using an efficient sub-pixel convolutional neural network. In Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR), Las Vegas, NV, USA, 27–30 June 2016; pp. 1874–1883. [CrossRef]
45. Balley, S.; Baella, B.; Christophe, S.; Pla, M.; Regnauld, N.; Stoter, J. Map specifications and user requirements. In *Abstracting Geographic Information in a Data Rich World: Methodologies and Applications of Map Generalization*; Springer International Publishing: Cham, Switzerland, 2014; pp. 17–52. [CrossRef]
46. Wang, Z.; Bovik, A.C.; Sheikh, H.R.; Simoncelli, E.P. Image quality assessment: From error visibility to structural similarity. *IEEE Trans. Image Process.* **2004**, *13*, 600–612. [CrossRef] [PubMed]
47. Che, Y.; Li, X.; Liu, X.; Wang, Y.; Liao, W.; Zheng, X.; Zhang, X.; Xu, X.; Shi, Q.; Zhu, J.; et al. 3D-GloBFP: The first global three-dimensional building footprint dataset. *Earth Syst. Sci. Data* **2024**, *16*, 5357–5374. [CrossRef]
48. Cheng, B.; Girshick, R.; Dollar, P.; Berg, A.C.; Kirillov, A. Boundary IoU: Improving object-centric image segmentation evaluation. In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), Nashville, TN, USA, 20–25 June 2021; pp. 15334–15342. [CrossRef]
49. Zhang, Z.; Liu, Q.; Wang, Y. Road extraction by deep Residual U-Net. *IEEE Geosci. Remote Sens. Lett.* **2018**, *15*, 749–753. [CrossRef]
50. Isola, P.; Zhu, J.; Zhou, T.; Efros, A.A. Image-to-image translation with conditional adversarial networks. In Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR), Honolulu, HI, USA, 21–26 July 2017; pp. 1125–1134. [CrossRef]

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